

Differential Equations

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1 Fundamentals of Differential Equations

1.1 What are Differential Equations?

- A **differential equation** (DE) is an equation which expresses the relationship between expressions which are expressed using derivatives.

For example, Newton's Second Law, $\vec{F} = m\vec{a}$, is a differential equation since acceleration $\vec{a}$ is the time-derivative of velocity ($\vec{a} = \frac{d\vec{v}}{dt}$) and the second time-derivative of position ($\vec{a} = \frac{d^2\vec{x}}{dt^2}$):

$$\vec{F} = m\vec{a} = m\frac{d\vec{v}}{dt} = m\frac{d^2\vec{x}}{dt^2}$$

- The solutions to differential equations are *functions*:
 - The **general solution** to a differential equation represents all possible functions that satisfy the equation without initial conditions.
For example, the general solution to $\frac{dy}{dt} = y$ is $y = Ce^t$, where C can be any constant
 - If **initial/boundary conditions** are given, we may be able to determine a **particular solution** which is a form of the general solution that also satisfies the conditions.
For example, if given the condition $y(0) = 10$, then the particular solution to $\frac{dy}{dt} = y$ is $10e^t$
 - An **initial value problem (IVP)** is a differential equation constrained by as many **initial conditions** (known derivatives at one given time) needed to allow us to achieve a particular solution. The number of initial conditions required is the order of the differential equation.
ex: $f(0) = 0, f'(0) = 1, f''(0) = 5$ are a set of conditions for a second-order IVP.
 - A **boundary value problem (BVP)** is a differential equation constrained by as many **boundary conditions** (known values at many different times) as needed. The number of boundary conditions required is also equal to the order of the DE.
ex: $f(0) = 0, f(1) = 10, f(2) = 5$ are a set of conditions for a second-order BVP.
 - The **interval of validity** for the solution to an IVP or BVP is the largest possible continuous interval on which the solution is defined and satisfies the conditions.
 - **Explicit solutions** are written in “function form”, where the dependent variable is isolated completely ($y = f(t)$)

Implicit solutions is any solution not written explicitly (e.g. $y^2 = \dots$ or $x^2 + y^2 = \dots$)

1.2 Classifying Differential Equations

Differential equations can be classified according to a number of properties:

Ordinary vs. Partial

- A differential equation is **ordinary** (an *ordinary differential equation* or *ODE*) if there are no partial derivatives anywhere in the function (i.e., we only deal with one independent variable)
ex: $\frac{dy}{dt} + t^2y = 0$ is an ODE because the only independent variable is t .

- A differential equation is **partial** (a *partial differential equation* or *PDE*) if it contains partial derivatives (i.e. we deal with multiple independent variables)
ex: $\frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$ is a PDE because there are multiple independent variables, x and t .

Order

The **order** of a differential equation is the order of the highest-order derivative present anywhere in the equation.

ex: The order of $\frac{dy}{dx} + y^{10} = x$ is 1, and the order of $y^2 \frac{d^3 y}{dx^3} + x \frac{d^2 y}{dx^2} = 0$ is 3.

Linear

A differential equation is **linear** if all derivatives of the dependent variable (y , $\frac{dy}{dx}$, $\frac{d^2 y}{dx^2}$, etc.) are represented linearly. That is, we can represent the DE in the form:

$$a_0(x)y + a_1(x)y' + a_2(x)y'' + \dots + a_n(x)y^{(n)} = f(x),$$

where $a_n(x)$ and $f(x)$ are arbitrary functions of x .

Examples

- $\cos(x) \frac{d^2 y}{dx^2} + \frac{dy}{dx} = 2x^2 y$ is linear
- $\left(\frac{dy}{dx}\right)^2 + y^2 = 0$ is nonlinear because $\left(\frac{dy}{dx}\right)^2$ and y^2 are nonlinear in $\frac{dy}{dx}$ and y .
- $\frac{dy}{dx} + \cos(y) = 0$ is nonlinear because $\cos(y)$ is nonlinear in y .

Degree

The **degree** of a differential equation is the polynomial degree of the highest-order derivative of the dependent variable.

This means that all linear differential equations are of the first-degree!

Examples

- $y' + y^2 = 1$ is first-degree because the highest-order derivative, y' , is first-degree.
- $(y')^2 + y^2 = 1$ is second-degree because the highest-order derivative, y' , is squared.
- $\left(\frac{d^2 y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + y^2 = x^6$ is third-degree because the highest-order derivative, $\frac{d^2 y}{dx^2}$, is cubed.

Autonomous

An **autonomous** differential equation is only governed by the current state of the variables. That is, it only contains the dependent variable or function.

More generally, for first-order DEs, autonomous differential equations take the form:

$$\frac{dy}{dx} = f(y).$$

If x appears in an autonomous DE, it should appear as a function of y .

Higher-order autonomous DEs take the form $a_0(y)y + a_1(y)y' + a_2(y)y'' + \dots + a_n(y)y^{(n)} = f(y)$.

Examples

- An exponential function can be represented by the DE $\frac{dy}{dx} = ky$. It is autonomous because its rate of change at any x is *only* governed by the value of $y(x)$.
- The DE which models a logistic function is also autonomous: $\frac{dP}{dt} = kP(L - P)$ where k, P are constants. The rate at which the population is growing at any time *only* depends on the population and not what time it is.
- The DE $y'(t) = t^2$ is *not* autonomous because the rate of change can vary based on the time t .

Homogeneous

A differential equation is **homogeneous** if it is linear AND it has no terms that *only* contain either the independent variable or a constant. They may, however, be coefficients of the derivatives of the dependent variable.

In other words, a homogeneous DE takes the form:

$$a_0(x) + a_1(x)y' + a_2(x)y'' + \dots + a_n(x)y^{(n)} = 0.$$

Examples

- The DE $\frac{dy}{dt} = 2xy$ is homogeneous because it is both linear, and there are no terms which only contain x .
- The DE $\frac{dy}{dt} = 2xy + 1$ is *not* homogeneous because there is the constant term 1.

1.3 Solving by Antidifferentiation

- The most fundamental differential equation relates some **rate of change** of a dependent variable with respect to an independent variable to another function.
- If we say $f'(t) = 2t$, we are saying that the rate of change of f at a time t is $2t$.
 - In other words, we are saying that the derivative of f is $2t$, so to find f , we take the antiderivative of $2t$ with respect to t :

$$\begin{aligned}f'(t) &= 2t \\ \int f'(t) dt &= \int 2t dt \\ f(t) + C_0 &= t^2 + C_1\end{aligned}$$

- Since we took two indefinite integrals, we ended up with two constants of integration. Usually, we implicitly drop one of the constants to simplify the work:

$$f(t) = t^2 + C.$$

where $C = C_1 - C_0$

- The DE above is *first-order* since we only work with f' in the equation. If it was made into an IVP, we would only need one constraint:
 - Let's add the initial value $f(0) = 4$. Then we can find a particular solution by plugging into f :

$$f(0) = 0^2 + C = 4 \implies C = 4$$

$$f(t) = t^2 + 4.$$

- Higher-order DEs may be represented in the form $f^{(n)}(t) = \dots$
 - Our constant of integration would become an $(n - 1)^{\text{th}}$ -degree polynomial as we integrate n times.
 - An IVP for a DE of that form would require at least n boundary conditions.

Example: $f'''(t) = 24t^3 + 6t, f(0) = 2, f'(0) = 5, f''(0) = 10$

Find the general solution:

$$\int f'''(t) dt = \int (24t^3 + 6t) dt$$

$$\int f''(t) dt = \int (6t^4 + 3t^2 + C_0) dt$$

$$\int f'(t) dt = \int \left(\frac{6}{5}t^5 + t^3 + C_0t + C_1 \right) dt$$

$$f(t) = \frac{1}{5}t^6 + \frac{1}{4}t^4 + C_0t^2 + C_1t + C_2$$

Solve for C_0, C_1, C_2 :

$$f(0) = \dots + C_2 = 2 \implies C_2 = 2$$

$$f'(0) = \dots + C_1 + \dots = 5 \implies C_1 = 5$$

$$f''(0) = \dots + 2C_0 + \dots = 10 \implies C_0 = 5$$

Write the particular solution:

$$f(t) = \frac{1}{5}t^6 + \frac{1}{4}t^4 + 5t^2 + 5t + 2.$$

1.4 Intervals of Validity

1.5 Existence and Uniqueness Theorems

- Often, when considering an IVP, we must first decide whether a solution exists to the IVP at all (existence), and whether if a solution to the IVP is the *only* solution (uniqueness).
- For example, the IVP $\frac{dy}{dx} = y^{\frac{2}{3}}, y(0) = 0$ can have two solutions: $y = x^3$ and $y = 0$.

Existence Theorem

Existence Theorem for First Order IVPs

Assume the initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$. If we select a rectangular interval R on $f(x, y)$ containing (x_0, y_0) , where $R = (a, b) \times (c, d)$, and $f(x, y)$ is **continuous** along R , then there must exist **at least** one solution to the IVP, $y(x)$, which is valid along a subinterval of (a, b) , $(x_0 - h, x_0 + h)$.

In simpler terms, let us have a DE in the form $\frac{dy}{dx} = f(x, y)$, and we want to find a solution passing through the point (x_0, y_0) .

Consider all pairs of values (x, y) in the intervals $x \in (a, b)$ and $y \in (c, d)$ which create a rectangular region R . If $f(x, y)$ is continuous throughout these intervals, and (x_0, y_0) is a point contained in R , then the existence theorem holds, and there must exist at least one function $y(x)$ that solves the initial value problem.

The solution $y(x)$ will have an interval of validity that is a subset of (a, b) , denoted $(x_0 - h, x_0 + h)$.

Uniqueness Theorem

Uniqueness Theorem for First Order IVPs

Let an IVP $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$ satisfy the existence theorem over $R = (a, b) \times (c, d)$. Then if $\frac{\partial f}{\partial y}$ is **continuous** over R , there must exist **exactly** one solution to the IVP, $y(x)$, which is valid along a subinterval of (a, b) .

Assume we have an IVP that already satisfies the existence theorem involving the DE in the form $\frac{dy}{dx} = f(x, y)$, meaning $f(x, y)$ is continuous over some interval R containing (x_0, y_0) . If $\frac{\partial f}{\partial y}$ is *also continuous* over R , then the uniqueness theorem holds, and there exists *exactly* one solution for the IVP.

1.6 Analyzing Differential Equations

Recall the the following form of a first-order DE:

$$\frac{dy}{dx} = f(x, y).$$

The function f in this case is known as the **slope function**, where the slope of the tangent line at $(x, y(x))$ on any solution curve to the DE.

We can sketch a **slope field** which represents

Analyzing Autonomous DEs

There are even simpler ways to represent an autonomous DE. Recall that an autonomous DE is one where the derivative only depends on the dependent variable:

$$\frac{dy}{dx} = f(y).$$

Phase Portraits and Asymptotic Stability

Since an autonomous DE $\frac{dy}{dx} = f(y)$ *only* depends on y , we can draw a one-dimensional “number line” to analyze the DE. This number line is known as a **phase portrait**. Where $\frac{dy}{dx}$ is positive, draw an arrow

2 First-Order Differential Equations

2.1 Separable First-Order ODEs

A separable first-order differential equation is in the form:

$$f(y)\frac{dy}{dx} = g(x).$$

This is the most simple type of differential equation to solve. *Separate* all functions and differentials in y to one side of the equation and all functions and differentials in x to the other, then integrate both sides:

$$\int f(y) dy = \int g(x) dx$$

Finally, solve for y in terms of x .

Example: $\frac{dy}{dx} = 6y^2x$

1. Separate

$$\frac{1}{y^2} dy = 6x dx$$

2. Integrate

$$\begin{aligned}\int \frac{1}{y^2} dy &= \int 6x dx \\ -\frac{1}{y} &= 3x^2 + C\end{aligned}$$

3. Solve for y

$$y = -\frac{1}{3x^2 + C} = \frac{1}{C - 3x^2}$$

Loss of a Solution

Let us solve the separable DE $\frac{dy}{dx} = g(x)h(y)$, and let $h(r) = 0$, where r is a constant. Then, $y = r$ is a solution to the DE as it makes both sides 0.

However, when we separate, we will get the DE $\frac{dy}{h(y)}$, where we would implicitly have to assume $h(y) \neq 0$, causing us to “lose” the solution $y = r$. Thus, if a separable DE is in the aforementioned form and does not satisfy the existence and uniqueness theorem, we must check if there exists a r such that $y = r$ is a solution to the DE.

2.2 Linear First-Order ODEs

A linear first-order differential equation is in the form:

$$\frac{dy}{dx} + P(x)y = Q(x).$$

These types of equations are typically solved using an **integrating factor**. This is some function $\mu(x)$ which is multiplied to both sides of the equation:

$$\mu(x)\frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x)$$

But why is this useful? In our original equation, notice we see both y and its derivative in:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Then, recall that the product rule would imply that for some function $f(x)$,

$$f(x)\frac{dy}{dx} + f'(x)y = \frac{d}{dx}f(x)y.$$

However, since we lack some $f(x)$ in front of our $\frac{dy}{dx}$ term, we use $\mu(x)$, the *integrating factor*, to help us get into the product rule form and find a general solution:

$$\begin{aligned}\mu(x)\frac{dy}{dx} + \mu(x)P(x)y &= \mu(x)Q(x) \\ \frac{d}{dx}\mu(x)y &= \mu(x)Q(x) \\ \mu(x)y &= \int \mu(x)Q(x) dx \\ y &= \frac{1}{\mu(x)} \int \mu(x)Q(x) dx.\end{aligned}$$

Note that we are only dealing with the general case. When working with an actual equation, we must still remember the constant of integration: $y = \frac{1}{\mu(x)}(\int \mu(x)Q(x) dx + C)$.

We can solve for $\mu(x)$ by recalling the assumptions made by the product rule:

$$\begin{aligned}\cancel{\mu(x)\frac{dy}{dx}} + \mu(x)P(x)y &= \cancel{\mu(x)\frac{dy}{dx}} + \mu'(x)y = \frac{d}{dx}\mu(x)y \\ \mu(x)P(x)y &= \mu'(x)y\end{aligned}$$

From this, we get $\mu'(x) = \mu(x)P(x)$. We can now solve for the integrating factor:

$$\begin{aligned}\mu'(x) &= \mu(x)P(x) \\ \frac{\mu'(x)}{\mu(x)} &= P(x) \\ \int \frac{\mu'(x)}{\mu(x)} dx &= \int P(x) dx \\ \ln(\mu(x)) &= \int P(x) dx \\ \mu(x) &= e^{\int P(x) dx}\end{aligned}$$

So, the expanded general solution to a linear ODE in the form $\frac{dy}{dx} + P(x)y = Q(x)$ is:

$$y = \frac{1}{e^{\int P(x) dx}} \int e^{\int P(x) dx} Q(x) dx.$$

First-Order Linear Differential Equation

The general solution to the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ is:

$$y = \frac{1}{\mu(x)} \int \mu(x) Q(x) dx + \frac{C}{\mu(x)}$$

where $\mu(x) = e^{\int P(x) dx}$, the integrating factor.

Example: $x \frac{dy}{dx} + 2y = x^2 - x + 1$

First, rewrite the DE in the form $\frac{dy}{dx} + P(x)y = Q(x)$ by dividing both sides by x :

$$\frac{dy}{dx} + \frac{2}{x}y = x - 1 + \frac{1}{x}$$

We see that $P(x) = \frac{2}{x}$ and $Q(x) = x - 1 + \frac{1}{x}$. Find the integrating factor:

$$\mu(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2.$$

Now, multiplying the equation by $\mu(x)$ on both sides:

$$x^2 \frac{dy}{dx} + 2xy = x^3 - x^2 + x$$

$$\frac{d}{dx}(x^2 y) = x^3 - x^2 + x$$

$$x^2 y = \int (x^3 - x^2 + x) dx$$

$$x^2 y = \frac{1}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2}x^2 + C$$

$$y = \frac{1}{4}x^2 - \frac{1}{3}x + \frac{1}{2} + \frac{C}{x^2}.$$

2.3 Exact First-Order ODEs

We can rearrange certain differential equations to be in the following form:

$$f(x, y) dx + g(x, y) dy = 0.$$

Rearranging, we see that many DEs can be represented in the above form:

$$g(x, y) dy = -f(x, y) dx$$

$$\frac{dy}{dx} = -\frac{f(x, y)}{g(x, y)}$$

$$f(x, y) + g(x, y) \frac{dy}{dx} = 0$$

Exact differential equations are DEs derived from implicit differentiation.

For example, here is the exact DE derived from the implicit solution $x^2y^4 + y^3 = 8$:

$$x^2y^4 + y^3 = 8$$

$$2xy^4 + 4x^2y^3 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (4x^2y^3 + 3y^2) = -2xy^4$$

$$2xy^4 dx + (4x^2y^3 + 3y^2) dy = 0$$

So, if implicit differentiation takes us from an implicit solution to an exact DE, how do we go from an exact DE to an implicit solution?

Conveniently, if we represent x as independent of y and set the function $F(x, y)$ equal to the solution, its **total differential** is the same as the differential equation:

$$\text{Let } F(x, y) = x^2y^4 + y^3 = 8$$

$$dF = \underbrace{2xy^4 dx}_{\frac{\partial F}{\partial x}} + \underbrace{(4x^2y^3 + 3y^2) dy}_{\frac{\partial F}{\partial y}}$$

Recall: the total differential dF of a function $F(x, y)$ is $dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$.

In other words, if $F(x, y) = 0$ is a solution to the DE, then the exact DE we began with represents the total differential of F , dF . We use the implied equality for $\frac{\partial F}{\partial x}$ to find an expression for $F(x, y)$:

$$\frac{\partial F}{\partial x} = 2xy^4$$

$$\int \frac{\partial F}{\partial x} dx = \int 2xy^4 dx$$

$$F = x^2y^4 + c(y)$$

Any function of y which we call $c(y)$ will vanish when computing $\frac{\partial F}{\partial x}$.

We can then use the equality for $\frac{\partial F}{\partial y}$ to solve for $c(y)$ and thus find a solution for $F(x, y)$:

$$F = x^2y^4 + c(y) \implies \frac{\partial F}{\partial y} = 4x^2y^3 + g'(y) = 4x^2y^3 + 3y^2$$

$$g'(y) = 3y^2 \implies c(y) = \int 3y^2 dy = y^3 + C$$

$$\implies F(x, y) = x^2y^4 + y^3 + C.$$

We can then write the implicit solution to the differential equation as $F(x, y) = 0$:

$$\begin{aligned}
 2xy^4 dx + (4x^2y^3 + 3y^2) dy &\Rightarrow x^2y^4 + y^3 + C = 0 \\
 &\Rightarrow x^2y^4 + y^3 = C.
 \end{aligned}$$

Is my DE exact?

Generally, we call a differential equation in the form $f(x, y) dx + g(x, y) dy = 0$ **exact** if there exists a function $F(x, y)$ such that $dF = f(x, y) dx + g(x, y) dy$.

We can check if any DE is exact without actually knowing F by using the symmetry of second derivatives, which states that:

$$\frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y} \right).$$

Since the total differential implies that $\frac{\partial F}{\partial x} = f(x, y)$ and $\frac{\partial F}{\partial y} = g(x, y)$, then we arrive at the following definition:

Exact Differential Equation

The differential equation $f(x, y) dx + g(x, y) dy$ is **exact** if:

$$\frac{\partial}{\partial y} f(x, y) = \frac{\partial}{\partial x} g(x, y).$$

Example: $(\cos x + x \cos y) \frac{dy}{dx} = y \sin x - \sin y$

First, rearrange into $f(x, y) dx + g(x, y) dy = 0$ form:

$$\begin{aligned}(\cos x + x \cos y) dy &= (y \sin x - \sin y) dx \\-(y \sin x - \sin y) dx + (\cos x + x \cos y) dy &= 0 \\(\sin y - y \sin x) dx + (\cos x + x \cos y) dy &= 0.\end{aligned}$$

Next, check if our DE is an exact DE:

$$\begin{aligned}\frac{\partial}{\partial y}(\sin y - y \sin x) &\stackrel{?}{=} \frac{\partial}{\partial x}(\cos x + x \cos y) \\ \cos y - \sin x &\stackrel{?}{=} -\sin x + \cos y \\ \cos y - \sin x &\stackrel{\checkmark}{=} \cos y - \sin x\end{aligned}$$

Now that we know our DE is exact, we can assume the DE has a solution $F(x, y) = C$.

Taking the total differential of both sides:

$$\begin{aligned}dF &= \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0 \\ &= (\sin y - y \sin x) dx + (\cos x + x \cos y) dy = 0\end{aligned}$$

We can then find F using either $\frac{\partial F}{\partial x}$ or $\frac{\partial F}{\partial y}$. Let's use $\frac{\partial F}{\partial x}$:

$$\begin{aligned}\frac{\partial F}{\partial x} &= \sin y - y \sin x \\\int \frac{\partial F}{\partial x} dx &= \int (\sin y - y \sin x) dx \\ F &= x \sin y + y \cos x + c(y)\end{aligned}$$

We can resolve $c(y)$ by using what we know about $\frac{\partial F}{\partial y}$:

$$\begin{aligned}\frac{\partial F}{\partial y} &= x \cos y + \cos x + c'(y) = \cos x + x \cos y \\ c'(y) &= 0 \implies c(y) = C_0\end{aligned}$$

Finally, write the solution:

$$\begin{aligned}x \sin y + y \cos x + C_0 &= C \\ x \sin y + y \cos x &= C\end{aligned}$$

Inexact Differential Equations

Sometimes, a differential equation can be "almost exact", i.e. it can be made exact with an *integrating factor* μ .

For a differential equation in the form $f(x, y) dx + g(x, y) dy = 0$:

- **Case 1:** if we can find a $\mu(x)$ such that $\frac{\partial}{\partial y} \mu(x)f(x, y) = \frac{\partial}{\partial x} \mu(x)g(x, y)$, then the differential equation $\mu(x)f(x, y) dx + \mu(x)g(x, y) dy = 0$ is exact.

$$\begin{aligned}
\frac{\partial}{\partial y}\mu(x)f(x,y) &= \frac{\partial}{\partial x}\mu(x)g(x,y) \\
\mu(x)\frac{\partial}{\partial y}f(x,y) &= \mu'(x)g(x,y) + \mu(x)\frac{\partial}{\partial x}g(x,y) \\
\mu(x)\left(\frac{\partial}{\partial y}f(x,y) - \frac{\partial}{\partial x}g(x,y)\right) &= \mu'(x)g(x,y) \\
\frac{\frac{\partial}{\partial y}f(x,y) - \frac{\partial}{\partial x}g(x,y)}{g(x,y)} &= \frac{\mu'(x)}{\mu(x)} \\
\int \frac{\frac{\partial}{\partial y}f(x,y) - \frac{\partial}{\partial x}g(x,y)}{g(x,y)} dx &= \int \frac{\mu'(x)}{\mu(x)} dx \\
\int \frac{\frac{\partial}{\partial y}f(x,y) - \frac{\partial}{\partial x}g(x,y)}{g(x,y)} dx &= \ln|\mu(x)| + C \\
e^{\int \frac{\frac{\partial}{\partial y}f(x,y) - \frac{\partial}{\partial x}g(x,y)}{g(x,y)} dx} &= \mu(x).
\end{aligned}$$

- **Case 2:** if we can find a $\mu(y)$ such that $\frac{\partial}{\partial y}\mu(y)f(x,y) = \frac{\partial}{\partial x}\mu(y)g(x,y)$, then the differential equation $\mu(y)f(x,y) dx + \mu(y)g(x,y) dy = 0$ is exact.

$$\begin{aligned}
\frac{\partial}{\partial y}\mu(y)f(x,y) &= \frac{\partial}{\partial x}\mu(y)g(x,y) \\
\mu'(y)f(x,y) + \mu(y)\frac{\partial}{\partial y}f(x,y) &= \mu(y)\frac{\partial}{\partial x}g(x,y) \\
&\vdots \\
e^{\int \frac{\frac{\partial}{\partial x}g(x,y) - \frac{\partial}{\partial y}f(x,y)}{f(x,y)} dy} &= \mu(y).
\end{aligned}$$

Note

When computing the integrating factor, **make sure it is a function of only the correct variable**. For example, $\mu(x) = 2y$ would be a contradiction: μ cannot be a function of y .

If computing both $\mu(x)$ and $\mu(y)$ result in a contradiction, then our differential equation cannot be solved using this integrating factor.

Example: $(4 \sin y + \frac{6}{x}) dx + x \cos y dy = 0$

First, check if this DE is exact:

$$\frac{\partial}{\partial y} \left(4 \sin y + \frac{6}{x} \right) \stackrel{?}{=} \frac{\partial}{\partial x} x \cos y$$
$$4 \cos y \neq \cos y.$$

Since our DE is not exact, we will check if an integrating factor $\mu(x)$ exists.

Compute the integrating factor:

$$\begin{aligned} \mu(x) &= e^{\int \frac{\frac{\partial}{\partial y} (4 \sin y + \frac{6}{x}) - \frac{\partial}{\partial x} x \cos y}{x \cos y} dx} \\ &= e^{\int \frac{4 \cos y - \cos y}{x \cos y} dx} \\ &= e^{\int \frac{3}{x} dx} = e^{3 \ln x} = x^3. \end{aligned}$$

Since x^3 is a function of only x , $\mu(x)$ is a valid integrating factor.

Multiply both sides of the DE by x^3 to make it exact, then solve:

$$(4x^3 \sin y + 6x^2) dx + x^4 \cos y dy = 0$$

$$\begin{aligned} \frac{\partial F}{\partial x} &= 4x^3 \sin y + 6x^2 \implies F = \int (4x^3 \sin y + 6x^2) dx \\ &= x^4 \sin y + 2x^3 + c(y) \end{aligned}$$

$$\begin{aligned} \frac{\partial F}{\partial y} &= x^4 \cos y + c'(y) = x^4 \cos y \\ \implies c'(y) &= 0 \implies c(y) = C \\ \implies F &= x^4 \sin y + 2x^3 + C \end{aligned}$$

$$x^4 \sin y + 2x^3 = C.$$

2.4 Homogeneous First-Order ODEs

Homogeneous Function

A function $f(x_1, x_2, \dots, x_n)$ is **homogeneous** of order k (where $k \in \mathbb{Z}$) if for any t ,

$$f(tx_1, tx_2, \dots, tx_n) = t^k f(x_1, x_2, \dots, x_n).$$

For example, $f(x, y) = y^2 + xy$ is a homogeneous function of the second order because $f(tx, ty) = (ty)^2 + (tx)(ty) = t^2 y^2 + t^2 xy = t^2(y^2 + xy) = t^2 f(x, y)$.

A first-order differential equation is **homogeneous** if it can be written in the form:

$$f(x, y) dx = g(x, y) dy$$

where f and g are homogeneous functions of the same order.

This is different than the definition of “homogeneous” for linear first-order ODEs.

Conveniently, as we will see, all homogeneous first-order DEs can be solved using a simple substitution. If we rearrange the differential equation, we get:

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)}.$$

We can use the fact that f and g are homogeneous of the same order to see that:

$$\frac{f(tx, ty)}{g(tx, ty)} = \frac{t^k f(x, y)}{t^k g(x, y)}.$$

Thus, if we let $t = \frac{1}{x}$, we see that $\frac{f(x, y)}{g(x, y)}$ can be represented in terms of $\frac{y}{x}$:

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)} = \frac{f(1, \frac{y}{x})}{g(1, \frac{y}{x})} = h\left(\frac{y}{x}\right)$$

In fact, an easier way to check if a differential equation in the form $\frac{dy}{dx} = \dots$ is homogeneous is to check if the RHS can be written in as a function of $\frac{y}{x}$. In other words, all differential equations of the form $\frac{dy}{dx} = h\left(\frac{y}{x}\right)$ are homogeneous.

Next, we can use the substitution $u = \frac{y}{x}$ to help us simplify the differential equation:

$$\frac{dy}{dx} = h\left(\frac{y}{x}\right) = h(u)$$

We can write $\frac{dy}{dx}$ in terms of u and $\frac{du}{dx}$ using the product rule:

$$\begin{aligned} u &= \frac{y}{x} \implies y = ux \\ \therefore \frac{dy}{dx} &= \frac{du}{dx}x + u \frac{dx}{dx} = \frac{du}{dx}x + u \end{aligned}$$

This way, our differential equation becomes separable:

$$\begin{aligned} \frac{du}{dx}x + u &= h(u) \\ \frac{du}{dx}x &= h(u) - u \\ \frac{du}{h(u) - u} &= \frac{dx}{x} \end{aligned}$$

We can then solve for u via integration and then back-substitute $u = \frac{y}{x}$.

Example: $\frac{dy}{dx} = \frac{y^3 + x^2y}{xy^2 + x^3}$

First, realize that the DE is indeed homogeneous because we can rearrange it into:

$$(y^3 + x^2y) dx = (xy^2 + x^3) dy,$$

and both $y^3 + x^2y$ and $xy^2 + x^3$ are homogeneous of order 3.

Make the substitution $u = \frac{y}{x} \implies y = ux \implies \frac{dy}{dx} = \frac{du}{dx}x + u$:

$$\begin{aligned} \frac{du}{dx}x + u &= \frac{u^3x^3 + x^3u}{x^3u^2 + x^3} \\ &= \frac{x^3(u^3 + u)}{x^3(u^2 + 1)} \end{aligned}$$

Conveniently, $\frac{u^3+u}{u^2+1} = u$, so our DE becomes extremely simple:

$$\begin{aligned} \frac{du}{dx}x + u &= u \\ \frac{du}{dx}x &= 0 \end{aligned}$$

Assuming $x \neq 0$, then $\frac{du}{dx}$ must be 0 and thus u must be a constant:

$$\frac{du}{dx} = 0 \implies u = C$$

Back-substitute $u = \frac{y}{x}$:

$$\frac{y}{x} = C \implies y = Cx.$$

2.5 Bernoulli Differential Equations

A **Bernoulli** differential equation is *almost* a first-order linear DE, as it is in the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n.$$

Since we have y^n , our differential equation is no longer linear and cannot be immediately solved using an integrating factor.

We can divide both sides by y^n to eliminate y from the RHS:

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

If we make the substitution $u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{y^n}{1-n} \frac{du}{dx}$, we can eliminate y from the equation entirely:

$$y^{-n} \frac{y^n}{1-n} \frac{du}{dx} + P(x)u = Q(x)$$

In fact, we can rearrange to see that our substitution has transformed our DE into a regular linear first-order DE which we can solve using an integrating factor:

$$\frac{1}{1-n} \frac{du}{dx} + P(x)u = Q(x)$$

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x).$$

Given the integrating factor $\mu(x) = e^{\int (1-n)P(x) dx}$, we get an implicit solution:

$$u = y^{1-n} = \frac{1-n}{\mu(x)} \int \mu(x)Q(x) dx.$$

2.6 Numerical Methods

Euler's Method

Newton's Method

3 Higher-Order Differential Equations

3.1 Introduction to Second-Order Differential Equations

Second-Order Initial Value Problems

- A **second-order IVP** is an initial value problem involving a second-order differential equation.
- These require at least two initial conditions (or two boundary conditions):
 $y(x_0) = \alpha, y'(x_0) = \beta.$

Existence and Uniqueness for Higher-Order Linear ODEs

Recall that an n th-order linear ODE can be written in the form:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x).$$

Let us subject it to the initial conditions $y(x_0) = y_0, y'(x_0) = y'_0, \dots, y^{(n)}(x_0) = y_0^{(n)}.$

The **existence and uniqueness theorem for an n th-order linear ODEs** requires that:

- all coefficient functions $a_i(x)$ and $f(x)$ are continuous on a finite open interval I containing x_0
- the leading coefficient function $a_n(x)$ is never 0 on that interval I

If both conditions are met, then the theorem guarantees that there must exist a solution $y(x)$ which satisfies the IVP on the open interval I .

Existence and Uniqueness Theorem for Higher-Order ODEs

Consider an n th-order linear ODE in the form:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x).$$

Let this ODE be subject to the initial conditions $y(x_0) = c_0, y'(x_0) = c_1, \dots, y^{(n)}(x_0) = c_n$, where x_0 and $\{c_i\}$ are constants.

Then if all $a_i(x)$ are continuous on a finite open interval I where $x_0 \in I$, and the leading coefficient function $a_n(x) \neq 0$ for all $x \in I$, then the **existence and uniqueness theorem** asserts that there *must* exist a solution $y(x)$ which satisfies the IVP, and this solution is the *only* solution.

Second-Order Linear ODEs

A **second-order linear ODE** is one which can be written in the form:

$$a(x)y'' + b(x)y' + c(x)y = f(x).$$

If $f(x) = 0$, the ODE is also **homogeneous** (i.e. we can write it as $a(x)y'' + b(x)y' + c(x)y = 0$).

Principle of Superposition

Principle of Superposition for Homogeneous DEs

If y_1 and y_2 are both solutions to the same **linear and homogeneous** differential equation, then $C_1y_1 + C_2y_2$ must also be a solution to the differential equation $\forall C_1, C_2 \in \mathbb{R}$.

Fundamental Sets of Solutions

- A **fundamental solution set** of a DE is the smallest set of solutions where the most general solution is a linear combination of all solutions in the set.

In other words, it is the smallest set of solutions where after applying the principle of superposition, we achieve the *most general* solution to the DE.

- A **linear and homogeneous** n th-order ODE will have a fundamental solution set with **exactly** n solutions.

Fundamental Solution Sets

Let $Y = \{y_i\}$ be a set of solutions which satisfy a differential equation. Y is a **fundamental solution set** if:

1. $\nexists y_i, y_j \in Y$ s.t. $y_i = cy_j$ where c is a constant
2. $\sum_i c_i y_i$ is the most general solution to the differential equation, where $\{c_i\}$ is an arbitrary sequence of constant coefficients

The Wronskian

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Validating Solution Sets

Let $Y = \{y_i\}$ be a set of solutions which satisfy a differential equation.

If $W(y_1, y_2, \dots, y_n) \neq 0$, then Y is a **fundamental solution set** to the differential equation.

3.2 Homogeneous Second-Order ODEs with Constant Coefficients

- If $r = \lambda_1, \lambda_2$, then $y = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$
- If $r = \lambda$, then $y = C_1 e^{\lambda t} + C_2 \lambda e^{\lambda t}$
- If $r = \alpha + \beta i$, then $y = e^{\alpha t}(C_1 \cos(\beta t) + C_2 \sin(\beta t))$

3.3 Nonhomogeneous Second-Order ODEs: Undetermined Coefficients

3.4 Nonhomogeneous Second-Order ODEs: Variation of Parameters

3.5 Nonconstant Coefficients: Reduction of Order

A second order linear homogeneous ODE with nonconstant coefficients is in the form:

$$y'' + P(x)y' + Q(x)y = 0.$$

If we have a second-order linear homogeneous ODE *in standard form* with a known solution $y_1(x)$ and the coefficient $P(x)$, we can find a second solution $y_2(x)$ using the method of **reduction of order**. This is the analog of the integrating factor for second-order ODEs.

We assume a solution of the form:

$$y_2(x) = v(x)y_1(x)$$

Thus the derivatives of y_2 are:

$$\begin{aligned}y_2'(x) &= v'(x)y_1(x) + v(x)y_1'(x) \\y_2''(x) &= v''(x)y_1(x) + 2v'(x)y_1'(x) + v(x)y_1''(x).\end{aligned}$$

3.6 Euler Differential Equations

3.7 Power Series Solutions

4 Modeling Differential Equations

4.1 First-Order Linear Models

Population Dynamics

Radioactive Decay

Salt Mixing Problems

4.2 Second-Order Linear Models

Spring-Mass Systems

Electrical Circuits

4.3 Nonlinear Models

Predator-Prey Models

Chemical Reactions

5 Laplace Transforms

Laplace Transform

Let $f(t)$ be function in the time domain t . Then the Laplace transform of $f(t)$ is:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

6 Systems of Differential Equations

6.1 Matrix Methods for Systems of DEs

Writing Systems of DEs in Matrix Form

6.2 Numerical Methods

7 Fourier Series

7.1 Deriving the Fourier Series

Periodic & Orthogonal Functions

Periodic Function

A function $f(x)$ is **periodic** with period T if $f(x + T) = f(x)$ for all x .

ex: $\sin(\omega x)$ and $\cos(\omega x)$ are periodic with period $T = \frac{2\pi}{\omega}$.

Orthogonal Functions

If the functions $f(x)$ and $g(x)$ are non-zero, they are **orthogonal** for $a \leq x \leq b$ if:

$$\int_a^b f(x)g(x) dx = 0.$$

Fourier Transform

$$\mathcal{F}\{f(t)\} = \hat{f}(\xi) = \int_0^\infty e^{-2\pi i \xi t} f(t) dt$$